

GASTOLENS

Technical Product Overview

React 18 · Vite 5 · Supabase · PostgreSQL · Tailwind CSS · PWA

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Product Summary

GastoLens is a production-deployed personal finance PWA that removes the friction of expense logging. Users describe transactions in plain language — *"dinner with friends 45 euros"* — and the built-in NLP engine automatically extracts the amount, currency, and category. Voice input via the Web Speech API is also supported. The app is live at `gastolens.vercel.app`, fully authenticated, backed by Supabase PostgreSQL with Row Level Security, and synchronises in real time across all devices with zero infrastructure cost (free tiers).

Frontend

React 18	Hooks, Context API, JSX
Vite 5	Build tool, HMR, asset pipeline
Tailwind 3	Utility-first, neon palette
Recharts 2	Donut + daily line charts
Lucide React	Consistent icon library

Backend & Infrastructure

Supabase	PostgreSQL + Auth + Realtime
Vercel	Deploy, CDN global, HTTPS
Service Worker	PWA cache v3, offline support
Web Speech API	Voice recognition (es-ES)

Application Architecture

UI Layer	React components: *App, Navigation, BalanceCards, ExpenseInput, Charts, ExpenseHistory, ExpenseDetail, AuthScreen*
Logic Layer	Custom hooks — `useExpenses` (CRUD + realtime sync), `useAuth` (JWT session management), `useSpeech` (voice transcription)
State Layer	Context API: `AuthContext` wraps the Supabase session and exposes `user / login / register / logout` globally
NLP Layer	Pure JavaScript parser: numeric normalisation → amount extraction → keyword category scoring
Data Layer	`@supabase/supabase-js` client, environment-variable configured, zero secrets committed to source
Cloud Layer	Supabase Cloud: PostgreSQL 15, GoTrue Auth, Realtime WebSocket engine

Database Schema — `public.expenses`

Column	Type	Notes
`id`	`uuid PK`	`gen_random_uuid()`
`user_id`	`uuid FK`	references `auth.users`
`amount`	`decimal(10,2)`	stored in EUR after conversion
`original_amount`	`decimal(10,2)`	amount in source currency
`currency`	`text`	`EUR` / `RON` / `USD`
`category`	`text`	NLP-assigned main category
`subcategory`	`text`	one of 40+ detected subcategories
`description`	`text`	NLP-generated readable label
`raw_text`	`text`	original unmodified user input
`date`	`timestamptz`	expense timestamp

RLS Policy: `USING (auth.uid() = user_id)` — each user can only read and write their own rows.

NLP Engine

The core differentiator of GastoLens is a custom, fully client-side NLP parser written in vanilla JavaScript. It operates on free-form Spanish text in three sequential phases:

1. **Numeric normalisation** — converts Spanish number words to digits. *"treinta y cinco"* → 35. Handles compounds up to the thousands and mixed forms (*"dos mil cuatrocientos cincuenta y tres"* → 2453).
2. **Amount extraction** — multi-layer regex pipeline detects: `45.00`, `45 euros`, `1.200,50`, slang currency (*pavos, cholos, leus*) and multi-currency symbols (EUR, RON, USD, $). Exchange rates applied automatically on match.
3. **Category scoring** — each token scores against a keyword matrix across all 8 categories (40+ subcategories). The category with highest cumulative score wins. Ties are resolved by token position weight.

Input (raw text)	Amount	Currency	Category / Subcategory
"cena con amigos 45 euros"	45.00	EUR	Ocio / Restaurantes
"supermercado cuarenta euros"	40.00	EUR	Alimentacion / Supermercado
"taxi 12 leus"	12.00	RON	Transporte / Taxi
"Netflix quince dolares"	15.00	USD	Ocio / Streaming
"treinta y cinco pavos de cena"	35.00	EUR	Ocio / Restaurantes
"farmacia ocho euros"	8.00	EUR	Salud / Farmacia

Feature Inventory

Feature	Implementation details
Natural language input	Custom NLP parser, 40+ keyword rules, slang support
Voice input	Web Speech API, `es-ES` locale, real-time transcript
Auto-categorisation	8 categories, 40+ subcategories, keyword scoring matrix
Multi-currency	EUR, RON, USD; fixed exchange rates; both amounts stored
Balance cards	Today / Week / Month totals, computed on every state update
Donut chart	Category breakdown, interactive tooltip with category colour
Line chart	Daily spending trend for current month
Expense history	Full paginated list, category filter, detail modal per entry
Real-time sync	Supabase Realtime `postgres_changes` subscription
Authentication	Email + password, confirmation email, JWT auto-refresh
User profile	Display name independent of email, account creation date
Offline support	Service Worker v3, cache-first for hashed assets
PWA shell	Safe-area (iPhone Dynamic Island), home screen icon
Error handling	DB error banner, toast notifications, graceful fallback

Security Model

Row Level Security (RLS)

Enabled on `public.expenses`. Supabase enforces `auth.uid() = user_id` at the database level. No application code can bypass this policy — even a compromised frontend cannot read another user's data.

JWT + Session Management

Supabase GoTrue issues signed JWTs on login. Tokens auto-refresh before expiry via `onAuthStateChange` listener. Sessions persist in `localStorage` with automatic recovery on page load.

Transport Security & API Isolation

All client–server traffic uses HTTPS (enforced by Vercel and Supabase). The Service Worker is configured to *never* cache cross-origin requests — Supabase API calls always bypass the SW and go directly to the network, preventing stale data from being returned. Supabase credentials are stored exclusively as environment variables, absent from the source repository.

Market Opportunity

The personal finance app market is valued at $1.57B (2023, CAGR 5.7%). Mint, the dominant free tracker, was shut down in January 2024, leaving millions of users actively seeking alternatives. The Spanish-speaking market (500M+ native speakers) has no dominant dedicated expense tracker: existing competitors either require bank credentials — which European users widely distrust — or lack natural language input entirely.

Competitor	Price	Market	Gap vs GastoLens
YNAB	$109/yr	USA only	No NLP, no voice, steep learning curve
Copilot	$95/yr	USA only	iOS-only, US bank connection required
Monarch Money	$99/yr	USA only	Requires bank connection, no NLP
Fintonic	Freemium	Spain	Requires bank credentials, no NLP
Mint	—	USA	**Discontinued January 2024**

Monetisation Paths

1. **Subscription ($4.99–$9.99/month)** — Add Stripe or LemonSqueezy paywall. Backend is already cloud-based; tier enforcement is approximately one weekend of development.
2. **Freemium model** — Free up to 50 expenses/month; paid tier unlocks unlimited entries, CSV/PDF export, and budget goals (designed but not implemented).
3. **English NLP + global reach** — Extending the parser to English takes approximately one week, opening the product to the full global personal finance market.
4. **Native mobile app** — The React codebase wraps with Capacitor to produce iOS and Android binaries for App Store and Google Play distribution.
5. **White-label** — Sell the codebase to fintech companies or banks needing a branded expense tracker for their customers.

Assets Included in Sale

- ✔ Full source code (React 18 + Vite 5)
- ✔ Custom Spanish NLP engine (core IP)
- ✔ Supabase project transfer
- ✔ Vercel project transfer (CI/CD pipeline)
- ✔ All environment variable configuration
- ✔ Brand: name, colour palette, UI design system
- ✔ Service Worker (offline + PWA logic)
- ✔ Web Speech API integration
- ✔ SQL schema + RLS policy definitions
- ✔ Technical documentation (PDF)

Infrastructure at handover:

- Supabase free tier ($0/month)
- Vercel free tier ($0/month)
- No vendor lock-in
- Zero migration required

Not included:

- ✘ Paying users or revenue
- ✘ Custom domain (.com)
- ✘ Registered trademark
- ✘ App Store / Play Store listing